Genomer
Biyoteknoloji Arge San.ve Tic. Ltd. Şti.

Microbiome Testing Line

Clinical Product Portfolio

Comprehensive real-time PCR-based molecular diagnostics for quantitative microbiota assessment across gynecological, pediatric, and urological applications

Product Portfolio

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FEMOBIOME II LINE

Women's Urogenital Microbiota

- 3 test variants: FEMOBIOME II (16 tubes), PrimaScreen (8 tubes), SecundaScreen (8 tubes)
- Normobiota, anaerobes, aerobes, mycoplasmas, STIs, HPV, herpesviruses
- Vaginal and cervical sampling incl. self-collection
- Visual traffic light reporting system

ENTEROBIOME KIDS

Children's Gut Microbiota

- Covers 99.9% of prokaryotic mass in children's feces
- 6 bacterial phyla + Euryarchaeota + Candida
- Pathogenicity markers: mecA, srr2, tcdA, tcdB
- Normocenosis/dysbiosis status with diversity estimation

ANDROBIOME

Male Urogenital Microbiota

- 2 variants: Androbiome (full) and Androbiome Screen
- No analogues worldwide
- Normal, opportunistic microbiota and STI detection
- 6 biomaterial types supported

All tests use real-time PCR technology with quantitative results | From sample collection to report interpretation

FEMOBIOME II LINE — Women's Microbiota Assessment

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Expanded line of tests for comprehensive women's urogenital microbiota profiling with real-time PCR. Features optimized pathogen panels, quantitative GBS assessment for pregnancy, and viral detection including HPV and herpesviruses.

What's New in FEMOBIOME II

- Expanded and optimized microorganism detection panel
- More informative normobiota evaluation across age groups
- Quantitative Group B Streptococcus assessment for pregnancy planning
- Obligate pathogen control for asymptomatic infections
- HPV detection: types 16, 18, 45 + 11 additional high-risk types
- Herpesvirus detection: HSV-1, HSV-2, CMV

Three Test Variants

FEMOBIOME II

Universal test — 16 tubes. Most comprehensive capabilities for full microbiota profiling.

PrimaScreen

Shortened version — 8 tubes. Focused on bacterial vaginosis diagnostics.

SecundaScreen

Complex test — 8 tubes. Infectious and inflammatory urogenital disease diagnostics.

FEMOBIOME II — Detection Panel Comparison

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Comprehensive comparison of microorganisms detected by each test variant

Category	Organism / Marker	FB II	Prima	Secunda
Control	Human DNA / Total bacterial load	✓	✓	✓
Normobiota	Lactobacillus spp., L. crispatus, L. jensenii, L. gasseri, L. iners, Bifidobacterium	✓	Partial	Partial
Aerobes	Staphylococcus, Streptococcus, S. agalactiae, Enterobacteriaceae, Enterococcus, Haemophilus	✓	—	✓
Anaerobes	Gardnerella, Fannyhessea, Mobiluncus, Anaerococcus, Peptostreptococcus, Bacteroides/Prevotella, Sneathia, Megasphaera, BVAB1-3	✓	Key only	✓
Mycoplasmas	U. urealyticum, U. parvum, M. hominis	✓	✓	✓
Yeast	Candida spp., Candida albicans	✓	✓	✓
STIs	C. trachomatis, M. genitalium, N. gonorrhoeae, T. vaginalis	✓	—	✓
Herpes	HSV-1, HSV-2, CMV	✓	—	✓
HPV	HPV 16, 18, 45 + 11 high-risk types	✓	—	—

FEMOBIOME II provides the most clinically comprehensive profiling across all categories

FEMOBIOME II — Clinical Indications & Sampling

Clinical Indications

Infectious Diseases

Inflammatory symptoms → FEMOBIOME II; BV symptoms → FEMOBIOME II or SecundaScreen

Asymptomatic Cases

Preparation for pelvic surgery; Preventive screening → FEMOBIOME II or PrimaScreen

Treatment Monitoring

Post-treatment assessment of infectious diseases and BV

Obstetrics

Pregnancy planning and ART; Prophylactic exams during pregnancy; Infertility and miscarriage workup → FEMOBIOME II

Biomaterial & Preparation

Sample Types

- Vaginal swab (specialist or self-collected)
- Cervical swab in transport medium
- Compatible with liquid cytology media (PreservCyt, BD SurePath, EASYPREP, CellPrep)

Patient Preparation

- 24h: No syringing, tampons, antiseptics, transvaginal ultrasound
- 48h: No colposcopy
- 72h: No unprotected sexual contacts
- 2 weeks after antibiotic/probiotic therapy
- 4 weeks after STI therapy

FEMOBIOME II — Report Interpretation

Traffic Light Indicator System

State	Pathogens Detected	No Pathogens Viruses/Myco/Candida $\geq 10^4$	No Pathogens No Significant Findings
Eubiosis	RED	YELLOW	GREEN
Moderate Dysbiosis	RED	YELLOW	YELLOW
Severe Dysbiosis	RED	RED	RED

Mycoplasmas: U. urealyticum, U. parvum, M. hominis. Candida: C. spp., C. albicans. Thresholds: $\geq 10^4$ GE/ml.

Report Components

- Indicator icon — visual summary combining microbiota state and pathogen presence
- Linear histogram — proportions of normobiota, anaerobes, aerobes, and mycoplasmas with color-coded quantity levels
- Columnar histogram — detailed distribution of conditionally pathogenic microorganisms within each group
- Personalized comments and infographics for each patient

Result Categories

- Eubiosis — no pathogens, no significant viruses/mycoplasmas/Candida
- Warning — moderate dysbiosis or elevated viruses/mycoplasmas/Candida
- Alarm — severe dysbiosis or obligate pathogen detected

FEMOBIOME II — Sample Report

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✓ Eubiosis

Normal microbiota dominance

Conclusion

Microbiota status — eubiosis: normobiota dominance, Lactobacillus spp. relative proportion 99.8% (L. crispatus, L. iners, L. jensenii).

No pathogens detected.

Biomaterial Control Parameters

Human genomic DNA: 4.9 lg GE/ml (Ref: ≥3.5)

Total bacterial load: 6.2 lg GE/ml (Ref: ≥4.0)

Pathogens — All Not Detected

SEXUALLY TRANSMITTED INFECTIONS

Chlamydia trachomatis — not detected
Mycoplasma genitalium — not detected
Neisseria gonorrhoeae — not detected
Trichomonas vaginalis — not detected

HERPESVIRUSES

HSV-1 — not detected
HSV-2 — not detected
CMV — not detected

HUMAN PAPILLOMAVIRUSES (HPV)

HPV 16 — not detected
HPV 18 — not detected
HPV 45 — not detected
HPV 31/33/35/39/51/52/56/58/59/66/68 — not detected

Microbiota Composition

Example: Eubiosis case — vaginal swab

Normobiota 99.8%

Threshold: ≥80% normobiota = eubiosis

Microorganism	lg GE/ml	%
NORMOBIOTA		
Lactobacillus spp.	6.2	99.8%
Bifidobacterium spp.	1.7	<1%
AEROBES		
Staphylococcus spp.	—	—
Streptococcus spp.	—	—
Enterobacteriaceae	—	—
Enterococcus spp.	2.4	<1%
Haemophilus spp.	—	—
ANAEROBES		
Gardnerella vaginalis	—	—
Fannyhessea vaginae	—	—
Mobiluncus spp.	—	—
Anaerococcus spp.	2.9	<1%
Peptostreptococcus spp.	2.6	<1%
Bacteroides / Prevotella	3.3	<1%
Sneathia / Leptotrichia / Fusobacterium	—	—
Megasphaera / Veillonella / Dialister	2.4	<1%
BVAB1 / BVAB2 / BVAB3	—	—
MYCOPLASMAS		
Ureaplasma urealyticum	—	—
Ureaplasma parvum	<4.0	<1%
Mycoplasma hominis	—	—
YEAST		
Candida spp.	—	—
Candida albicans	—	—

FEMOBIOME II — Technical Specifications

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Kit Specifications

- Analysis time: from 2.5 hours (with sample pretreatment)
- PCR formats: Automated (384-well) and manual (96-well)
- Package S: 12 tests (24 for PrimaScreen/SecundaScreen)
- Package A / A-TL: 24 tests (48 for PrimaScreen/SecundaScreen)
- Storage: +2 to +8°C (packages S, A, A-TL)
- Storage: -18 to -22°C (TechnoTaq MAX polymerase)
- Shelf life: 12 months
- Transport medium: STOR-F, STOR-M

Compatible Platforms

Extraction Platforms

- Allsheng Auto-Pure 96
- KingFisher Flex (PREP-MB-RAPID II)

Liquid Cytology Media

- PreservCyt (Hologic Inc., USA)
- BD SurePath (Becton, Dickinson, USA)
- EASYPREP (YD Diagnostics, South Korea)
- CellPrep (CP Biodyne, South Korea)
- Cell Preservative Solution (Human Lituo, China)

Sample Pretreatment

- PREP-PK for scrapes in BD SurePath and CellPrep media

ENTEROBIOME KIDS — Children's Gut Microbiota

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Reagent kit for quantitative analysis of gut microbiota composition in children's stool by real-time PCR. Detects microbial taxa covering 99.9% of total prokaryotic mass.

Detection Scope

- Actinomycetota: Bifidobacterium spp. (child and adult species), Coriobacteriia
- Bacillota: Cl. leptum gr., F. prausnitzii, Lachnospiraceae, Lactobacillaceae, Streptococcus spp.
- Bacteroidota: Bacteroides, Prevotella, Parabacteroides, Alistipes, Butyricimonas spp.
- Verrucomicrobiota: Akkermansia muciniphila
- Pseudomonadota: Enterobacterales
- Euryarchaeota: Methanobacteriaceae
- Candida spp. (yeast fungi)

Key Capabilities

- Normocenosis/dysbiosis status determination
- Normal microbiota diversity estimation
- Bifidobacteria species-level identification (B. bifidum, B. longum, B. infantis, B. breve, B. adolescentis)
- Bacillota/Bacteroidota ratio calculation
- Pathogenicity and resistance markers: mecA (methicillin resistance), srr2 (S. agalactiae invasiveness), tcdA/tcdB (C. difficile toxins)

Kit Features

- Multiplex format with internal control
- Automatic results generation via RealTime_PCR software
- Preset templates for major PCR instruments
- Biomaterial: feces including meconium

ENTEROBIOME KIDS — Microbiome Development

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Four Phases of Gut Microbiota Formation in Children

Phase 1: Birth
Streptococcus spp., Enterobacterales

Phase 2: 14 Days
Bifidobacterium, Lactobacillaceae, Clostridium, Bacteroidota

Phase 3: Complementary Foods
Bacteroidota, Bacillota, Verrucomicrobiota, Actinomycetota

Phase 4: Weaning
Stabilization — adult-like composition

Factors Affecting Microbiota Formation

- **Type of birth:** Natural birth → stable composition; C-section → delayed colonization, more C. difficile, Enterococcus, Klebsiella, Veillonella
- **Feeding type:** Breastfed → Bifidobacterium dominant (B. breve, B. bifidum, B. longum); Formula-fed → Bacteroides and Clostridium dominant
- **Gestational age:** Preterm → delayed anaerobic colonization, higher Enterobacteriaceae and opportunistic organisms
- **Antibiotic use:** Reduces diversity, increases drug-resistant Enterobacteriaceae and Clostridium, decreases Bifidobacteriaceae

> **A species-rich gut ecosystem is more resilient to adverse stresses. Reduced diversity is associated with disease presence.**

ENTEROBIOME KIDS — Detailed Detection Panel

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Normal Microbiota

Actinomycetota

- Bifidobacterium spp. (child species: B. bifidum, B. longum, B. infantis, B. breve)
- Bifidobacterium spp. (adult species: B. adolescentis, B. catenulatum, B. animalis)
- Coriobacteriia

Bacillota

- Cl. leptum gr. (cluster IV)
- F. prausnitzii
- Lachnospiraceae (cluster XIVa)
- Lactobacillaceae
- Lactococcus lactis
- Streptococcus spp.
- Veillonellaceae
- Erysipelotrichaceae

Bacteroidota

- Bacteroides spp.
- Prevotella spp.
- Parabacteroides spp.
- Alistipes spp.
- Butyricimonas spp.

Other

- Akkermansia muciniphila (Verrucomicrobiota)
- Blautia spp. (acetogens)
- Desulfovibrio spp. (sulfate-reducing)
- Methanobacteriaceae (methanogens)

Opportunistic & Markers

Pathobionts

- C. difficile (tcdA, tcdB markers)
- C. perfringens gr.
- Enterobacterales
- Enterococcus spp.
- S. aureus (mecA marker)
- S. agalactiae (srr2 marker)

Fungi

- Candida spp.
- Candida albicans

ENTEROBIOME KIDS — Clinical Relevance

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Key Clinical Markers

- **Bacillota/Bacteroidota ratio:** Shift toward Bacteroidota may indicate inflammatory process
- **F. prausnitzii:** Anti-inflammatory biomarker. Promotes IL-10/IL-12 secretion, inhibits IL-8. Decreased in inflammatory bowel disease
- **Akkermansia muciniphila:** Main mucin metabolizer. Enhances barrier function. Produces propionate and acetate for butyrate production
- **Bifidobacterium species:** Child species (B. bifidum, B. longum, B. infantis, B. breve) ferment breast milk oligosaccharides, activate T-regulatory cells. Adult species ferment plant polysaccharides

Disease Associations & Markers

Pathogenicity Markers

- **mecA** — methicillin resistance in Staphylococcus; strains resistant to methicillin
- **srr2** — S. agalactiae invasiveness; promotes bloodstream and CNS invasion, causing meningitis
- **tcdA / tcdB** — C. difficile enterotoxins A and B; cause acute inflammation, intestinal barrier damage, fluid influx

Disease Associations

- IBD (UC, Crohn's): ↑ Bacteroides/Prevotella, ↓ F. prausnitzii
- Celiac disease: ↓ Lactobacillus/Bifidobacteria, ↑ pro-inflammatory Veillonellaceae
- Necrotizing enterocolitis (preterm): associated with C. perfringens and Enterobacterales
- C. difficile carriage: up to 61% in children <1 year; normally eliminated by ~2 years

ENTEROBIOME KIDS — Report Output

Result Form Components

1. Normocenosis / Dysbiosis Status

Determined by relative abundance of normobiota vs. opportunistic microbiota in TBM. Classification guides clinical decision-making.

Normocenosis | Moderate Dysbiosis | Severe Dysbiosis

2. Diversity Estimation

Normal microbiota diversity across 6 phyla: Actinomycetota, Bacillota, Bacteroidota, Pseudomonadota, Fusobacteriota, Verrucomicrobiota. Lower diversity correlates with disease risk.

3. Bacillota / Bacteroidota Ratio

Key gut balance indicator. Shift toward Bacteroidota may indicate inflammatory process. Associated with metabolic disorders, obesity, and IBD.

4. Quantitative Assessment

Bifidobacteria/lactobacilli counts with species-level identification. Opportunistic microbiota and *Candida* quantification. Includes *F. prausnitzii* and *Akkermansia muciniphila* as anti-inflammatory biomarkers.

Clinical Associations

IBD: ↑ Bacteroides/Prevotella, ↓ F. prausnitzii. Celiac: ↓ Lactobacillus/Bifidobacteria. NEC: C. perfringens, Enterobacterales. C. difficile carriage patterns tracked.

Pathogenicity Markers

mecA
Methicillin-resistant *Staphylococcus* detection

srr2
Str. agalactiae invasiveness; promotes meningitis

tcdA
C. difficile enterotoxin A

tcdB
C. difficile enterotoxin B (cytotoxin)

Automated Report Generation

Results form auto-generated with recommended Real-Time PCR instruments and RealTime_PCR software. Preset templates configure settings and calculate results automatically. No manual data entry required.

Report Includes

- Quantitative normobiota estimation by phylum
- Bifidobacteria species identification
- Opportunistic microbiota and *Candida* levels
- Antibiotic resistance markers (mecA)
- Virulence markers (srr2, tcdA, tcdB)
- Bacillota/Bacteroidota ratio calculation
- Overall dysbiosis classification

ANDROBIOME — Male Urogenital Microbiota

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Comprehensive real-time PCR-based assay for male urogenital tract microbiota assessment and STI detection. No analogues worldwide. Hands-on time from 3 hours.

Androbiome (Full)

Detailed microbiome structure assessment

- Normal flora: Staphylococcus, Streptococcus, Corynebacterium, Lactobacillus spp.
- BV-associated: Gardnerella, Megasphaera, Sneathia, Ureaplasma, Mycoplasma, Atopobium
- Anaerobes: Bacteroides, Anaerococcus, Peptostreptococcus, Eubacterium spp.
- Other OM: Haemophilus, Pseudomonas/Ralstonia/Burkholderia, Enterobacteriaceae/Enterococcus
- Yeast: Candida spp.
- STIs: C. trachomatis, M. genitalium, N. gonorrhoeae, T. vaginalis

Androbiome Screen

Screening and acute diagnostics

- Shortened panel for rapid screening
- Normal: Staphylococcus, Streptococcus, Corynebacterium
- Opportunistic: Gardnerella, Ureaplasma, Mycoplasma, Enterobacteriaceae/Enterococcus, Candida
- Full STI panel: C. trachomatis, M. genitalium, N. gonorrhoeae, T. vaginalis
- Ideal for acute urogenital tract diseases

When to Use Which

- Screen: Screening, differential diagnosis of acute forms
- Full: Chronic diseases, ineffective treatment, reproductive function, IVF preparation

ANDROBIOME — Detection Panel Comparison

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Complete comparison of organisms detected by Androbiome and Androbiome Screen

Category	Organism	Full	Screen
Normal	Staphylococcus spp., Streptococcus spp., Corynebacterium spp.	✓	✓
Transit	Lactobacillus spp.	✓	✓
BV-associated	Gardnerella, Megasphaera/Veillonella/Dialister, Sneathia/Leptotrichia/Fusobacterium, Atopobium cluster	✓	—
Mycoplasmas	Ureaplasma urealyticum, U. parvum, Mycoplasma hominis	✓	✓
Anaerobes	Bacteroides/Porphyromonas/Prevotella, Anaerococcus, Peptostreptococcus/Parvimonas, Eubacterium	✓	—
Other OM	Haemophilus spp., Pseudomonas/Ralstonia/Burkholderia, Enterobacteriaceae/Enterococcus	✓	Partial
Yeast	Candida spp.	✓	✓
STIs	C. trachomatis, M. genitalium, N. gonorrhoeae, T. vaginalis	✓	✓

[Androbiome Detection Panel Table]

Complementary tests: HSV1/HSV2/CMV, HPVQuant-21, BacScreen OM, FEMOBIOME (couple exam)

ANDROBIOME — Clinical Applications & Sampling

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Clinical Indications

- **Androbiome Screen:** Screening and differential diagnosis of acute urogenital diseases (urethritis, balanoposthitis)
- **Androbiome (Full):** Chronic forms, ineffective treatment, reproductive function assessment, IVF preparation
- **Couple examination:** Combined with FEMOBIOME for infertility workup

Preparation Requirements

- Sexual abstinence or barrier contraception for 3 days
- Avoid antiseptics and antibacterial soaps
- Avoid urinating 1.5-2 hours before urethral sampling
- Test can be prescribed during antibiotic therapy for monitoring

Biomaterial Options

- **Epithelial scrapes:** Glans penis, foreskin, urethra (urethritis, balanoposthitis)
- **Ejaculate:** Epididymitis, prostatitis, asymptomatic STIs, male infertility
- **Prostatic fluid:** After prostate massage (acute prostatitis: massage prohibited)
- **Residual urine post-massage:** Alternative to prostatic fluid collection
- **First-portion urine:** STI detection only (acute inflammation with pain)
- **Biopsy samples:** Prostatic tissue in sterile saline

> Storage: 2-8°C up to 24 hours. Frozen: -18 to -22°C up to 1 month. Multiple PCR tests can use a single sample.

ANDROBIOME — Results Interpretation

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4-Step Interpretation

- Assess obligate pathogens (STIs) and Candida spp. — reported even with insufficient biomaterial
- Control biomaterial compliance: Human DNA $>10^3$, TBM $>10^4$. Both below threshold → insufficient sample
- Assess balance: normal vs opportunistic microbiota. Normal = 2+ genera of commensals at TBM $>10^5$
- Generate automatic conclusion: pathogen presence, Candida status, microbiome structure

Transit Microbiota (Lactobacillus)

- Marker of female partner's microbiota in male tract
- If >10% of TBM → microbiome composition NOT analyzed
- Recommend: repeat with barrier contraception or 3-day abstinence

Key Indicators

Control Parameters

- Human DNA: shows amount of human genomic material in sample
- Total Bacterial Mass (TBM): total bacteria for proportion calculations
- Total Microorganisms Detected (TMD): mathematical parameter for group ratios
- Candida and T. vaginalis NOT included in TBM

Result Categories

- 🟢 Normal composition — commensals (Staphylococcus, Streptococcus, Corynebacterium) dominant
- 🟡 Moderate disturbance — decreased normal microbiota proportion
- 🔴 Severe disturbance — opportunistic predominance or STI detected

Couple Examination

Combined ANDROBIOME + FEMOBIOME for comprehensive partner evaluation in infertility workup

Technology Platform — Why Real-Time PCR?

Real-Time PCR vs. Alternative Methods

vs. Bacterial Culture

- 50-80% of gut microorganisms are difficult to culture
- No dependency on organism viability
- Detects anaerobes that culture methods miss
- Faster turnaround time
- More comprehensive coverage

vs. NGS Sequencing

- Results are clinically actionable without specialized interpretation
- More cost-effective for routine diagnostics
- Standardized and reproducible
- Suitable as routine clinical method
- No bioinformatics expertise required

Key Platform Advantages

- All-in-one: from sample collection to report interpretation
- Automated (384-well) and manual (96-well) PCR formats
- Fast nucleic acid extraction
- Multiplex format for efficiency
- Quantitative results with automatic interpretation

Technical Specifications & Platform Compatibility

Extraction & PCR Platforms

DNA Extraction

- Allsheng Auto-Pure 96
- KingFisher Flex (PREP-MB-RAPID II)
- PREP-MB MAX (for ejaculate)
- PREP-NA PLUS, PREP-GS PLUS (alternatives)

PCR Setup

- Automated 384-well format
- Manual 96-well format
- Preset templates for major real-time PCR instruments
- Automatic results generation via RealTime_PCR software

Storage & Logistics

Kit Storage

- +2 to +8°C: packages S, A, A-TL
- 18 to -22°C: TechnoTaq MAX polymerase (package A)
- Shelf life: 12 months for all kits

Sample Storage

- 2-8°C: up to 24 hours
- 18 to -22°C: up to 1 month (frozen)
- Transport media: STOR-F, STOR-M

Compatible Liquid Cytology Media

- PreservCyt (Hologic Inc., USA)
- BD SurePath (Becton, Dickinson, USA)
- EASYPREP (YD Diagnostics, South Korea)
- CellPrep (CP Biodyne, South Korea)

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Microbiome Testing Solutions

Comprehensive molecular diagnostics for clinical microbiota assessment

FEMOBIOME II
Women's urogenital microbiota

ENTEROBIOME KIDS
Children's gut microbiota

ANDROBIOME
Male urogenital microbiota

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